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Mitochondrial-Targeted Phytochemicals in Reproductive Aging: Mechanisms and Translational Potential

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ABSTRACT

Purpose: Reproductive aging is increasingly recognized as a major contributor to infertility in both women and men. Although traditionally linked to endocrine decline, emerging evidence suggests that mitochondrial dysfunction is a central mechanism underlying impaired gamete quality and reduced reproductive capacity. This review aimed to evaluate the role of mitochondrial deterioration in reproductive aging and to examine the therapeutic potential of mitochondrial-targeted phytochemicals.

Methods: A narrative review of experimental, translational, and clinical studies was conducted focusing on mitochondrial dynamics, oxidative stress, bioenergetic impairment, and phytochemical-based interventions in reproductive aging. Relevant evidence involving oocytes, spermatozoa, and the reproductive microenvironment was critically synthesized.

Results: Aging-associated mitochondrial dysfunction was consistently associated with impaired ATP production, excessive reactive oxygen species generation, mitochondrial DNA damage, and disrupted cellular homeostasis, leading to reduced oocyte competence, sperm motility, and embryonic development. Several phytochemicals, including resveratrol, quercetin, epigallocatechin gallate, astaxanthin, lycopene, berberine, and coenzyme Q10, demonstrated protective effects through modulation of SIRT1–PGC-1α, AMPK, Nrf2, and PINK1–Parkin signaling pathways.

Conclusions: Targeting mitochondrial bioenergetics with phytochemicals may provide a promising strategy for preserving fertility and extending reproductive lifespan, although further large-scale clinical validation is required.

1 | Introduction: Reframing Reproductive Aging as a Bioenergetic Disorder

Global socioeconomic and demographic transformations have precipitated a pronounced secular trend toward delayed parenthood in high- and middle-income nations alike. Epidemiological data from the United States indicate that mean maternal age at first birth increased from 24.9 years in the year 2000 to 26.6 years in 2016, with particularly substantial increases observed among women in their third and fourth decades of life [1]. These demographic shifts are further corroborated by population-register analyses demonstrating that advanced maternal age is independently associated with a broad spectrum of adverse pregnancy outcomes, including gestational hypertension, placental pathology, and chromosomal abnormalities in offspring [2].

Abbreviations: AMPK, AMP-activated protein kinase; ART, assisted reproductive technology; ATP, adenosine triphosphate; EGCG, epigallocatechin-3-gallate; ETC, electron transport chain; ICSI, intracytoplasmic sperm injection; IVF, in vitro fertilization; mtDNA, mitochondrial DNA; Nrf2, nuclear factor erythroid 2-related factor 2; OXPHOS, oxidative phosphorylation; PGC-1α, peroxisome proliferator-activated receptor gamma coactivator 1-alpha; PINK1, PTEN-induced kinase 1; ROS, reactive oxygen species; SIRT1, sirtuin 1; TPP^+, triphenylphosphonium; $\Delta\Psi m$, mitochondrial membrane potential.

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As women increasingly attempt conception at advanced reproductive ages, age-related infertility emerges as a preeminent biomedical and public health challenge [3, 4]. The dominant conceptual framework for reproductive aging has historically foregrounded hormonal alterations—particularly the progressive decline of anti-Müllerian hormone (AMH), diminished follicle-stimulating hormone (FSH) responsiveness, and attenuating estrogen production, as the primary determinants of reduced ovarian reserve and oocyte quality [5, 6]. While the clinical salience of these endocrine changes is indisputable, they may represent downstream sequelae of, or parallel phenomena to, a more fundamental cellular pathology. Converging lines of evidence now support the proposition that mitochondrial deterioration not only accompanies but may precede and drive many of the endocrine and structural alterations traditionally ascribed to reproductive senescence [7, 8].

Mitochondria, the principal intracellular producers of adenosine triphosphate (ATP) via oxidative phosphorylation (OXPHOS), are simultaneously the primary intracellular source of reactive oxygen species (ROS), regulators of cytoplasmic calcium homeostasis, arbiters of apoptotic commitment, and custodians of a distinct, maternally inherited genome [9, 10]. In reproductive cells, the bioenergetic and genomic requirements are particularly stringent: meiotic spindle assembly and chromosome segregation demand high and sustained ATP flux; fertilization and early embryogenesis depend upon the quantitative and qualitative integrity of oocyte mitochondria; and the efficiency of spermatozoal flagellar motility is directly constrained by mitochondrial OXPHOS capacity [7, 11]. Any deterioration in mitochondrial function therefore reverberates across multiple dimensions of reproductive competence simultaneously [12, 13].

The objective of this study is that reproductive aging is fundamentally a bioenergetic disorder rooted in progressive mitochondrial dysfunction, and that phytochemicals, biologically active secondary plant metabolites, offer a mechanistically coherent, multi-target pharmacological strategy for mitochondrial modulation and fertility preservation. The following sections address the bioenergetic architecture of gametes, the molecular hallmarks of mitochondrial aging, the pleiotropic mitochondrial effects of key phytochemical classes, emerging translational evidence from assisted reproductive technology (ART) settings supported by comprehensive evidence tables, and future research imperatives toward precision mitochondrial medicine in reproductive health.

2 | Gamete Bioenergetics: The Exceptional Mitochondrial Dependence of Reproductive Cells

2.1 | Oocyte Mitochondrial Architecture and Functional Imperatives

The mammalian oocyte occupies a unique position in cellular biology by virtue of its extraordinary mitochondrial endowment [12]. Quantitative estimates consistently situate the mitochondrial complement of the mature human oocyte at 100,000–600,000 organelles, a concentration orders of magnitude greater than that encountered in typical somatic cells, representing the exclusive mitochondrial reservoir that the resulting embryo will inherit and upon which it will depend until zygotic genome activation [14, 15].

This remarkable organellar density is not a biological epiphenomenon; rather, it reflects the extraordinary energetic demands imposed by meiotic progression, polar body extrusion, and the metabolic programming of the periimplantation embryo in the absence of active mitochondrial replication until the blastocyst stage [12]. The indispensability of mitochondrial ATP for oocyte competence is most starkly demonstrated by its role in meiotic spindle assembly and chromosomal segregation fidelity. The bipolar meiotic spindle is assembled and maintained through ATP-dependent molecular motor activity, and its architectural integrity is exquisitely sensitive to bioenergetic insufficiency [16, 17] (Figure 1).

Insufficient mitochondrial ATP disrupts spindle checkpoint kinase function, cohesin maintenance, and the enzymatic activity of chromosomal motor proteins, collectively increasing the probability of mis-segregation events and the consequent generation of aneuploid gametes, recognized as the predominant cause of age-related embryo developmental failure and pregnancy loss [13, 18]. Beyond spindle dynamics, the spatial redistribution of mitochondria within the oocyte cytoplasm during maturation constitutes an independent determinant of developmental competence. In functionally competent oocytes, mitochondria redistribute toward the spindle poles, cortical regions, and pericortical areas during maturation through ATP-dependent cytoskeletal remodeling. Disrupted mitochondrial distribution, manifesting as perinuclear aggregation or cytoplasmic clustering, is a reproducible morphological hallmark of aging oocytes and correlates with reduced fertilization rates and impaired developmental potential [16, 19]. Oocyte mitochondria further function as metabolic programmers of the emerging embryo, establishing the metabolic trajectory from which all subsequent embryonic cells derive [14].

2.2 | Mitochondrial Control of Spermatozoal Function

The midpiece of the spermatozoon is structurally defined by a helical sheath of tightly packed mitochondria encircling the axonemal core, positioned to maximize ATP delivery to the dynein ATPases that power flagellar beating and, consequently, progressive motility [11, 20]. Beyond ATP generation for motility, sperm mitochondria participate integrally in capacitation signaling, calcium homeostasis, localized ROS generation for zona pellucida penetration, and the maintenance of nuclear chromatin compaction through their influence on intracellular redox balance [21]. Mitochondrial membrane potential ($\Delta\Psi m$) in spermatozoa has emerged as a particularly informative biomarker of functional competence: robust correlations between high $\Delta\Psi m$ and progressive motility, zona pellucida binding capacity, and in vitro fertilization success have been documented across multiple investigative platforms [11].

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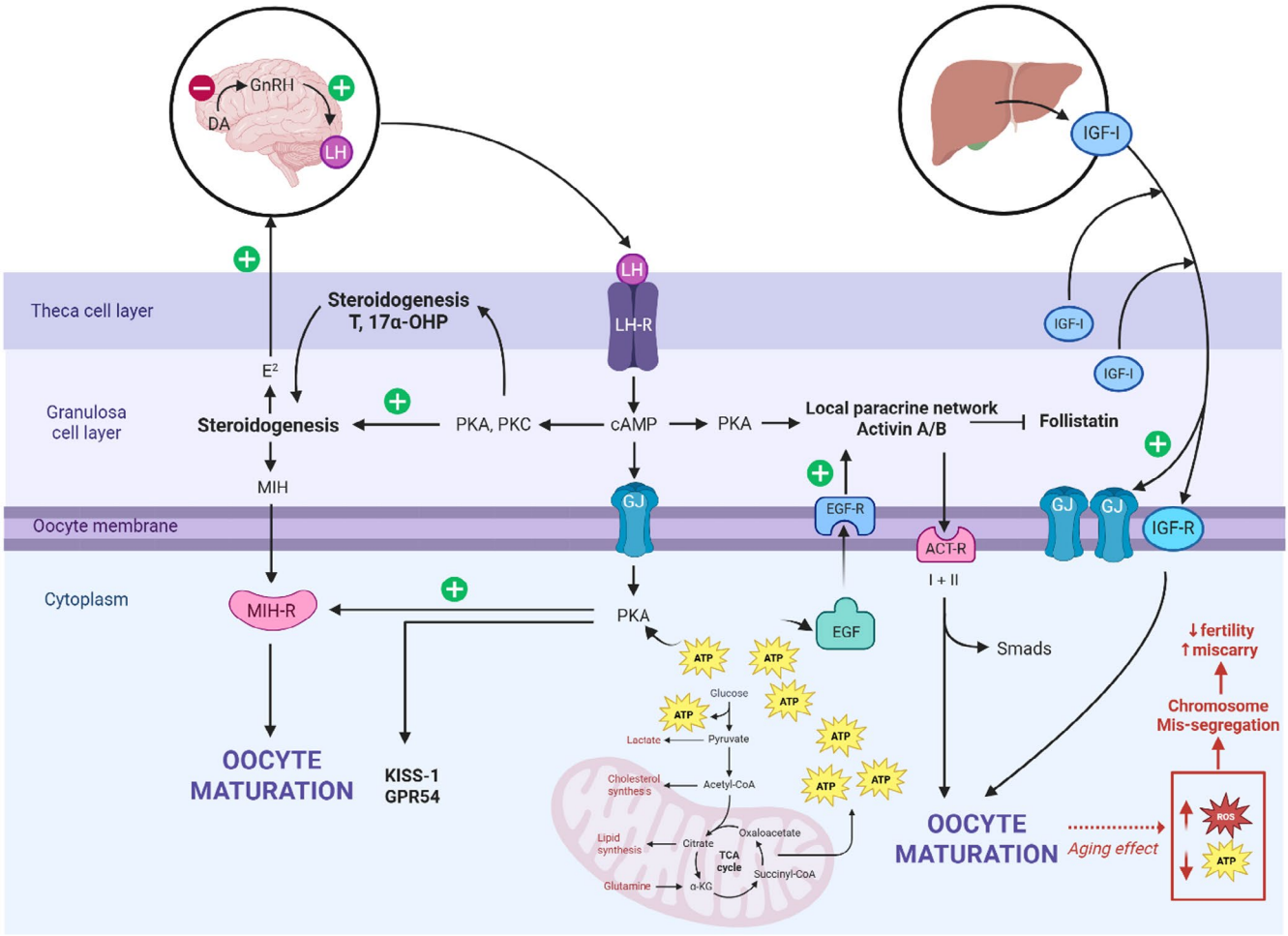


FIGURE 1 | Multifactorial signaling networks governing oocyte maturation. Oocyte maturation is regulated by coordinated endocrine, paracrine, and intracellular signals across the theca cell layer, granulosa cell layer, oocyte membrane, and cytoplasm. In the brain, GnRH stimulates (+) and dopamine (DA) inhibits (–) luteinizing hormone (LH) release. LH binds its receptor (LH-R) on granulosa cells, elevating cAMP and activating PKA and PKC, which drive steroidogenesis and gap junction (GJ)-mediated signaling into the oocyte. Intracellularly, PKA activates EGF signaling, which feeds back to support maturation. A local paracrine network involving Activin A/B and its antagonist Follistatin modulates EGF receptor (EGF-R) and Activin receptor (ACT-R I + II) signaling, converging on Smad transcription factors to promote oocyte maturation. Hepatic IGF-I acts via IGF receptors on granulosa cells and at gap junctions to further support maturation (+). Oocyte cytoplasmic maturation is supported by mitochondrial ATP production through the TCA cycle, fueled by glucose, pyruvate, glutamine, and acetyl-CoA, alongside lipid and cholesterol synthesis. KISS-1/GPR54 signaling and maturation-inducing hormone (MIH) acting on MIH-R also promote oocyte maturation. With aging, increased reactive oxygen species (ROS) and decreased ATP impair mitochondrial function, leading to chromosome mis-segregation, reduced fertility, and increased miscarriage risk (red panel).

2.3 | Mitochondrial Inheritance, the Bottleneck, and Developmental Programming

Mitochondria in mammals are transmitted almost exclusively through the maternal germline. Paternal mitochondria introduced by the fertilizing spermatozoon are selectively ubiquitinated and subsequently eliminated by the embryonic autophagic apparatus, a process critically dependent on the PINK1–Parkin pathway [9]. This stringent uniparental inheritance ensures that the quantitative and qualitative attributes of oocyte mitochondria exclusively determine the founding mitochondrial population of the embryo. The mitochondrial bottleneck, a marked reduction in the number of transmissible mitochondrial DNA (mtDNA) molecules during oogenesis, followed by massive amplification in the mature oocyte, is a fundamental determinant of mtDNA diversity in offspring [15]. These considerations underscore the singular importance of oocyte mitochondrial quality as a determinant not merely of fertilization success but of multigenerational health outcomes [13, 14].

3 | Molecular Signatures of Mitochondrial Aging in Reproductive Tissues

3.1 | Oxidative Stress and Mitochondrial ROS Generation

The electron transport chain (ETC), while constituting the primary cellular mechanism for ATP synthesis, is inherently imperfect: a finite fraction of electrons escapes at Complexes I and III and reacts with molecular oxygen to generate superoxide anion

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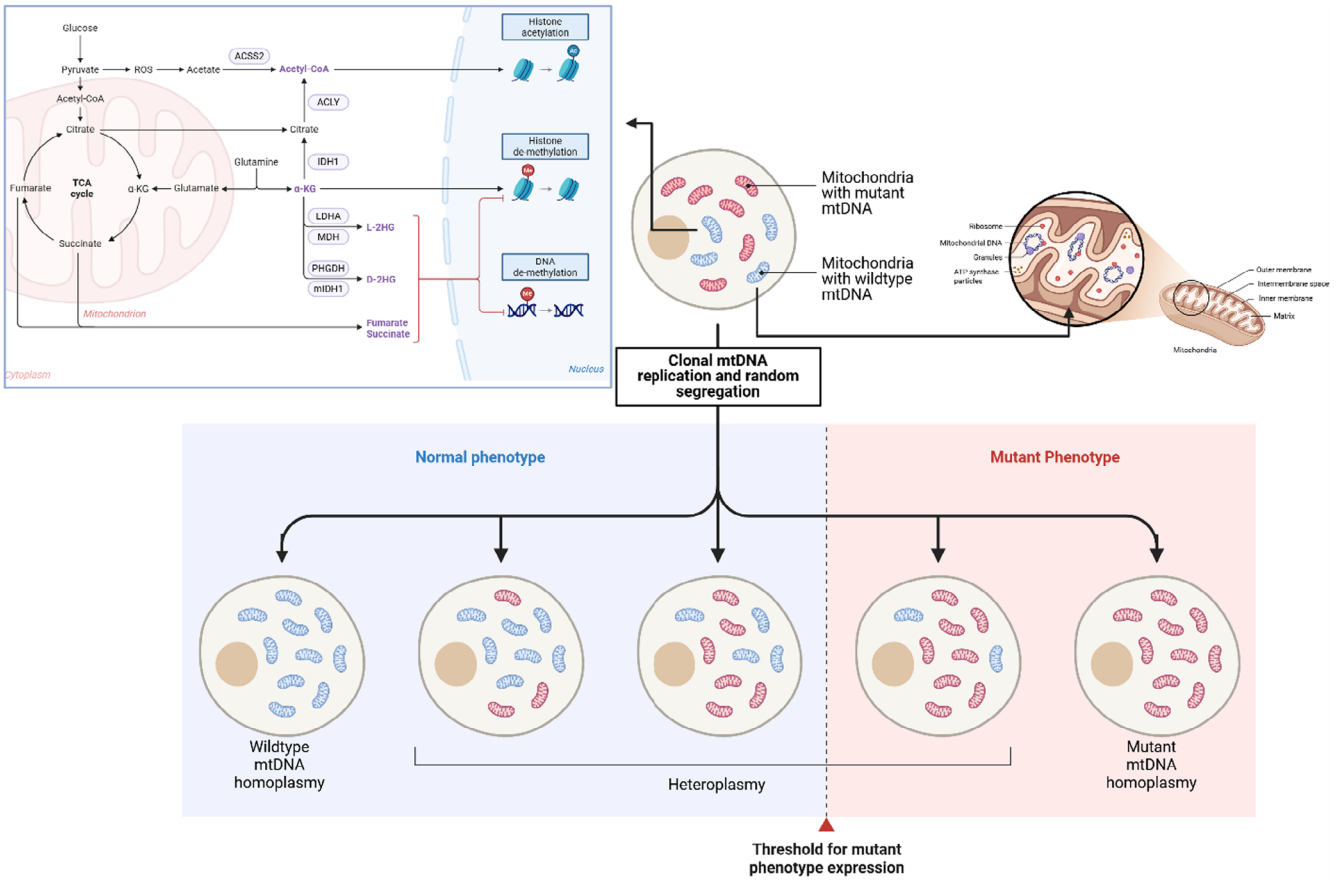


FIGURE 2 | Mitochondrial metabolic signaling, epigenetic regulation, and mtDNA segregation patterns relevant to reproductive aging. The figure illustrates two interconnected aspects of mitochondrial biology in the context of gamete quality and fertility. (Upper left) TCA cycle intermediates serve as critical mediators linking mitochondrial metabolic status to nuclear epigenetic regulation. Key metabolites such as acetyl-CoA and α-KG support histone acetylation and DNA/histone demethylation, respectively, promoting gene expression patterns favorable to normal gamete development. Conversely, accumulation of oncometabolites, L-2HG, D-2HG, fumarate, and succinate, as may occur under conditions of mitochondrial dysfunction and oxidative stress (ROS), suppresses epigenetic demethylation and may compromise oocyte and sperm developmental competence. (Upper right) Mitochondria within gametes exist as heterogeneous populations carrying either wildtype or mutant mtDNA. During replication and cell division, these mitochondria are randomly segregated into daughter cells. (Lower panel) This random segregation determines whether a cell achieves wildtype homoplasmy, heteroplasmy, or mutant homoplasmy. Critically, a mutant phenotype, characterized by impaired bioenergetics, reduced ATP output, and diminished fertility, is only manifested once the mutant mtDNA burden surpasses a defined threshold.

(O_2^{-}), the primary mitochondrial ROS precursor [17]. With advancing age, structural and compositional changes in ETC complexes, including oxidative modification of iron–sulfur clusters in Complex I, amplify superoxide production disproportionately to the antioxidant neutralization capacity [13, 22]. In reproductive cells, the consequences of this escalating oxidative burden are particularly severe. Gametes possess comparatively limited antioxidant defenses: oocytes exhibit age-dependent reductions in glutathione (GSH) biosynthesis, superoxide dismutase, catalase, and glutathione peroxidase activities [17]. The mitochondrial inner membrane, which is critically important for ATP synthesis, is highly enriched in polyunsaturated fatty acids, making it exquisitely susceptible to lipid peroxidation chain reactions that directly dissipate ΔΨm and impair OXPHOS efficiency [23]. The resultant self-amplifying cycle of mitochondrial ROS generation, membrane damage, and ETC dysfunction establishes a bioenergetic deterioration spiral that accelerates with reproductive age [6, 12] (Figure 2).

3.2 | Mitochondrial DNA Instability, Mutational Accumulation, and Heteroplasmy

The mitochondrial genome, a compact, 16.6-kilobase circular double-stranded DNA molecule, is uniquely vulnerable to oxidative attack due to its proximity to the ETC-derived ROS source, its lack of protective histones, and the limited base-excision repair capacity of the mitochondrial compartment [8, 12]. As a consequence, the mutation rate of mtDNA is estimated to be 10–17-fold higher than that of nuclear DNA in somatic tissues. A minimum copy number threshold, approximately 4000 copies per mature human oocyte, appears necessary for successful fertilization, while postimplantation development requires substantially more [13]. The concept of heteroplasmy, the coexistence within a single cell of both wildtype and mutant mtDNA molecules, is central to understanding the biological consequences of accumulated mutations. Above a cell-specific threshold, the progressive predominance

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of dysfunctional mitochondria precipitates a bioenergetic crisis [14, 15].

3.3 | Disruption of Mitochondrial Quality Control Systems

Mitochondrial homeostasis is maintained by an elaborate quality-control network encompassing mitochondrial biogenesis, the fusion–fission cycle, and mitophagy. PGC-1α, the master transcriptional coactivator of mitochondrial biogenesis, coordinates the expression of NRF1, NRF2, and TFAM to drive mtDNA replication and OXPHOS gene expression [10]. PGC-1α activity is positively regulated by both SIRT1, through NAD^+-dependent deacetylation, and AMPK, through phosphorylation at key serine/threonine residues [24]. In aging ovaries, the expression of SIRT1, SIRT3, and their downstream targets is consistently downregulated, resulting in hyperacetylation and inactivation of PGC-1α [13, 25]. Mitophagy, the selective elimination of dysfunctional mitochondria, is primarily executed through the PINK1–Parkin pathway. When ΔΨm collapses in a damaged mitochondrion, PINK1 stabilizes on the outer mitochondrial membrane and activates Parkin, which ubiquitinates outer membrane proteins and targets the damaged organelle for lysosomal degradation [9]. The age-dependent attenuation of PINK1–Parkin signaling efficiency in reproductive cells permits the accumulation of functionally deficient mitochondria that augment intracellular ROS without contributing productively to ATP synthesis. An aging-associated shift toward mitochondrial fragmentation further impairs the compensatory potential of the fusion apparatus [13, 19].

4 | Mitochondrial Dysfunction Across the Reproductive Axis

4.1 | Oocyte Aging and Mitochondrial Competence

The aging oocyte presents a pathognomonic constellation of mitochondrial abnormalities that mechanistically underpin its diminished developmental competence. Quantitative assessment of mitochondrial parameters in oocytes obtained from women and mice of advancing reproductive age consistently reveals progressive reductions in ATP content, ΔΨm, and mtDNA copy number, alongside escalating intracellular ROS concentrations [7, 16]. The seminal investigation by Pasquariello et al. [16] demonstrated that mitochondrial number and function are significantly correlated with spindle morphology and chromosome alignment, and that these bioenergetic parameters decline in a progressive and age-dependent fashion in both murine and human oocyte populations. This mechanistic chain, from mitochondrial bioenergetic insufficiency, through spindle assembly failure, to chromosomal mis-segregation and aneuploidy, constitutes the bioenergetic explanation for the well-documented exponential increase in oocyte aneuploidy rates with advancing maternal age [17, 18].

Oocyte aneuploidy is related to women with advanced age (> 38 years old) and poor responder. Poor responders have a tendency for low antral follicle count, number of oocytes, fertilization rate, embryo cleavage rate, pregnancy rate and birth rate. One of the etiologies of poor outcome in in vitro fertilization (IVF) in poor responder women is the unknown location of the meiotic spindle in the oocyte; thus, when intracytoplasmic sperm injection (ICSI) is performed, it can affect the spindle itself. Meiotic spindle view might improve the outcome of IVF in poor responder women with a significant improvement in cleavage rate and embryo quality [26]. Morphological oocyte abnormalities is also related to oocyte aneuploidy and correlated to aneuploidy embryo in IVF cycle. Oocyte quality is an important prognostic factor, since the nuclear and cytoplasmic maturity of the oocyte may be directly related to the success rate of ICSI. Although poor quality oocytes are difficult to fertilize, there is no correlation between ICSI fertilization success and oocyte dimorphism. Halim et al. [27] found that there was no significant difference in fertilization rates and embryo quality between normal morphology oocytes and oval-shaped oocytes [27].

4.2 | Mitochondrial Dysfunction in the Ovarian Microenvironment

The recognition that oocyte quality is substantially determined by the quality of the surrounding cellular microenvironment has expanded the conceptual scope of mitochondrial aging research to encompass the entire follicular unit. Granulosa cells (GCs) are critically dependent upon their own mitochondrial function for steroidogenesis, nutrient provisioning, and paracrine signaling [28]. Age-dependent reductions in SIRT1 and SIRT3 expression in GCs have been documented across multiple investigative platforms, resulting in attenuated PGC-1α activity, diminished mtDNA copy number, and reduced ATP content in these cells [24, 25]. Hypoxia at the follicular microvasculature exerts compounding suppressive effects on SIRT1 and mitochondrial biogenesis in GCs, creating a self-reinforcing cycle of follicular bioenergetic compromise [25].

4.3 | Mitochondrial Deterioration in Male Reproductive Aging

Although male reproductive aging proceeds less precipitously than the ovarian decline of the female, substantive age-related deterioration in sperm mitochondrial function has been rigorously documented. Spermatozoal ΔΨm progressively decreases with advancing paternal age, reflecting impaired ETC activity that arises from age-dependent accumulation of mitochondrial oxidative damage and large-scale mtDNA deletions [11, 21]. This ΔΨm decline is accompanied by disproportionate mitochondrial ROS generation, which initiates lipid peroxidation chain reactions in the highly unsaturated sperm plasma membrane, oxidizes flagellar axonemal proteins, and induces DNA strand breaks in the sperm chromatin [20, 29]. The functional ramifications for sperm reproductive competence include impaired progressive motility, elevated sperm DNA fragmentation indices, and reduced fertilization success.

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5 | Phytochemicals as Multi-Target Modulators of Mitochondrial Homeostasis

Phytochemicals, a structurally diverse group of bioactive secondary metabolites derived from plants, have attracted increasing attention as potential modulators of mitochondrial function in reproductive aging [30]. Unlike conventional single-target pharmacological agents, many phytochemicals exert broad regulatory effects across multiple interconnected cellular pathways, a characteristic that may be particularly relevant in the context of mitochondrial dysfunction, where oxidative stress, impaired bioenergetics, defective mitophagy, inflammatory signaling, and apoptotic activation develop in parallel rather than in isolation [10, 31]. This intrinsic polypharmacology provides a mechanistic basis for considering phytochemicals as candidate interventions capable of addressing the complex and multifactorial nature of reproductive aging.

In reproductive tissues, mitochondrial dysfunction extends beyond a simple decline in ATP production. In aging oocytes, impaired mitochondrial activity contributes to spindle instability, chromosomal segregation errors, abnormal calcium homeostasis, and reduced developmental competence [13, 16]. Similarly, mitochondrial deterioration within granulosa cells disrupts steroidogenesis and follicular metabolic support [25, 28], while in spermatozoa, mitochondrial impairment compromises flagellar motility, membrane integrity, and chromatin stability [11, 21]. Because these processes are tightly interconnected, interventions targeting only a single pathway may provide limited benefit. Phytochemicals capable of simultaneously modulating mitochondrial biogenesis, antioxidant defense systems, cellular metabolism, and mitochondrial quality-control pathways may therefore offer broader therapeutic flexibility in preserving reproductive function.

Several mechanistic pathways appear to underlie these mitochondria-protective effects. Many phytochemicals stimulate mitochondrial biogenesis through activation of the SIRT1–PGC-1α–NRF1–TFAM signaling axis, thereby promoting mtDNA replication, oxidative phosphorylation, and mitochondrial renewal [24, 32]. Others attenuate mitochondrial oxidative stress either through direct radical-scavenging activity or through activation of Nrf2/ARE-dependent antioxidant transcriptional programs [33, 34]. Additional compounds regulate mitochondrial quality control by enhancing PINK1–Parkin-mediated mitophagy [9, 35], while others influence cellular bioenergetics through AMPK-dependent metabolic reprogramming [36, 37]. Collectively, these effects converge on the preservation of mitochondrial homeostasis, maintenance of ATP production, stabilization of mitochondrial membrane potential, and reduction of apoptosis in reproductive cells.

The growing recognition of reproductive aging as a systems-level mitochondrial disorder has also stimulated interest in combination phytochemical strategies designed to achieve complementary bioenergetic effects. For example, compounds that enhance mitochondrial biogenesis may be combined with agents that stabilize mitochondrial membranes, improve electron transport chain efficiency, or suppress excessive ROS generation. Such combinatorial approaches may be particularly relevant in assisted reproductive settings, where mitochondrial dysfunction often arises from multiple overlapping mechanisms involving oxidative stress, metabolic insufficiency, and impaired mitochondrial quality control [31, 38]. As understanding of mitochondrial signaling networks continues to evolve, phytochemical-based modulation of reproductive bioenergetics is increasingly being viewed not simply as antioxidant supplementation, but as a broader strategy for mitochondrial restoration and fertility preservation.

6 | Key Classes of Phytochemicals Targeting Reproductive Mitochondria

6.1 | Polyphenols and Mitochondrial Biogenesis

Among the stilbenes, resveratrol (trans-3,5,4′-trihydroxystilbene) has accumulated the most extensive body of reproductive mitochondrial evidence. Resveratrol functions as a direct allosteric activator of SIRT1, binding to its N-terminal domain and lowering the Km for acetylated substrate through a mechanism definitively characterized by Borra et al. [32]. Activated SIRT1 deacetylates PGC-1α, relieving inhibitory acetylation and enabling PGC-1α–NRF1–TFAM-mediated induction of mitochondrial biogenesis, OXPHOS gene expression, and mtDNA replication [10, 25]. The summary can be seen in the Table 1.

The reproductive mitochondrial effects of resveratrol have been substantiated across multiple models. In aging mouse oocytes, short-term resveratrol administration significantly elevated ΔΨm and intracellular ATP content, improved implantation rates and live offspring rates, and upregulated ovarian Sirt1–7 isoform expression in a dose-dependent fashion [53]. The translational significance was substantially augmented by the rigorous randomized triple-blind placebo-controlled clinical trial of Mohamadhasani et al. [39], in which 800 mg/day of resveratrol administered to women with polycystic ovary syndrome (PCOS) undergoing IVF significantly increased PGC-1α and TFAM gene expression, elevated SIRT1 protein levels, improved mtDNA copy number and intracellular ATP content in GCs, reduced total oxidant status, and enhanced oocyte maturity rates and high-quality embryo rates compared with placebo. In the ovarian hypoxic microenvironment, resveratrol reverses $CoCl_2$-induced suppression of SIRT1, PGC-1α, and mitochondrial quantity in GC culture models [24, 25].

Quercetin (3,3′,4′,5,7-pentahydroxyflavone), a flavonol distributed abundantly in berries, onions, and broccoli, exerts complementary mitochondria-protective effects. Quercetin's principal mechanism involves SIRT3-dependent deacetylation of SOD2 at Lys68, substantially enhancing the catalytic dismutation of mitochondrial superoxide [40]. RNA-sequencing analyses of quercetin-treated aging oocytes revealed transcriptional signatures consistent with reduced mitochondrial ROS, suppressed apoptosis, and enhanced autophagy—mechanistic findings that translated into improved in vitro maturation (IVM) rates (+19.6%) and elevated blastocyst formation rates (+15.5%) in both aged murine and human oocyte experimental systems [40].

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TABLE 1 | Summary of phytochemical classes, primary mitochondrial targets, and mechanisms of action in reproductive cells.

Phytochemical class	Compound(s)	Primary mitochondrial target(s)	Key signaling pathway	Effect on reproductive mitochondria	Supporting evidence
Polyphenol (stilbene)	Resveratrol	SIRT1 (direct allosteric activator); AMPK co-activation	SIRT1 → PGC-1α deacetylation → NRF1/NRF2 → TFAM → mtDNA replication	↑ mtDNA copy number; ↑ ΔΨm; ↑ ATP synthesis; ↑ mitochondrial biogenesis; ↓ mitochondrial fragmentation	Borra et al. [32]; Nishigaki et al. [25]; Mohamadhasani et al. [39]
Polyphenol (flavonol)	Quercetin	SIRT3 (activates deacetylation of SOD2 at Lys68); SIRT1/2/3 preservation	SIRT3 → SOD2 activation → ↓ mitochondrial superoxide; preservation of SIRT1/2/3 → ↑ mitophagy fidelity	↓ mitochondrial ROS; ↑ ΔΨm; ↑ ATP; ↑ normal spindle assembly; ↓ apoptotic cascade initiation in oocytes	Cao et al. [40]; Jiao et al. [41]; Kim et al. [42]
Polyphenol (catechin)	EGCG (Epigallocatechin-3-gallate)	AMPK; Nrf2/ARE transcription factor; PINK1 upregulation	AMPK → PGC-1α; Nrf2 → antioxidant phase II enzymes (HO-1, NQO1); PINK1 → enhanced mitophagy initiation	↑ active mitochondria number; ↑ ATP; ↓ ROS; ↑ mitophagy efficiency; ↓ cytochrome c release; ↑ postovulatory oocyte viability	Zhang et al. [34]; Zhou et al. [35]; Huang et al. [43]
Carotenoid (xanthophyll)	Astaxanthin	Lipophilic ROS scavenging at inner mitochondrial membrane; Nrf2/ARE activation	Radical chain-breaking antioxidation spanning aqueous-lipid membrane interfaces; Nrf2 → SOD, CAT, GPx induction	↓ ETC-derived ROS; ↓ mitochondrial lipid peroxidation; ↑ ΔΨm in spermatozoa; ↑ sperm mitochondrial ATP; ↑ GPX4 expression	Eslami et al. [33]; Ghantabpour et al. [44]; Salman et al. [45]
Carotenoid (acyclic)	Lycopene	Singlet oxygen ($^1O_2$) quenching; lipid peroxyl radical interception within mitochondrial membranes	Nonenzymatic antioxidation via conjugated polyene electron delocalization; ↑ catalase and GPx expression	↓ sperm mitochondrial membrane lipid peroxidation; ↑ mitochondrial membrane integrity; ↓ ROS; ↑ antioxidant enzyme (SOD, CAT, GPx) activities	Durairajanayagam et al. [23]; Nouri et al. [46]; Babaei et al. [47]
Alkaloid (isoquinoline)	Berberine	Mitochondrial Complex I (partial reversible inhibition) → ↑ AMP:ATP → AMPK activation	AMPK → PGC-1α phosphorylation → ↑ biogenesis; AMPK → ACC inhibition → ↑ FAO; AMPK → ↓ mTORC1	↑ metabolic efficiency; ↑ sperm motility via restored ATP flux; ↓ mitochondrial ROS through NRF2; ↓ oxidative testicular damage	Turner et al. [37]; Song et al. [48]; Hassani-Bafrani et al. [49]

(Continues)

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TABLE 1 | (Continued)

Phytochemical class	Compound(s)	Primary mitochondrial target(s)	Key signaling pathway	Effect on reproductive mitochondria	Supporting evidence
Isoprenoid (ubiquinone)	Coenzyme Q10 (CoQ10)	ETC electron carrier (Complexes I/II→III); direct lipophilic ROS scavenger within inner mitochondrial membrane	Shuttle electrons through ETC → ↑ OXPHOS coupling; antioxidant (reduced form, ubiquinol) quenches lipid radicals	↑ ATP synthesis in oocytes; ↑ ovarian reserve; ↓ spindle abnormalities; ↓ aneuploidy; ↑ embryo quality; ↑ clinical pregnancy rate	Ben-Meir et al. [50]; Xu et al. [51]; Rodríguez-Varela & Labarta [52]

Abbreviations: ACC, acetyl-CoA carboxylase; AMPK, AMP-activated protein kinase; ARE, antioxidant response element; CAT, catalase; ETC, electron transport chain; FAO, fatty acid oxidation; GPx, glutathione peroxidase; mTORC1, mechanistic target of rapamycin complex 1; NRF, nuclear respiratory factor; Nrf2, nuclear factor erythroid 2-related factor 2; OXPHOS, oxidative phosphorylation; PGC-1α, peroxisome proliferator-activated receptor gamma coactivator 1-alpha; PINK1, PTEN-induced kinase 1; ROS, reactive oxygen species; SIRT, sirtuin; SOD, superoxide dismutase; TFAM, mitochondrial transcription factor A; ΔΨm, mitochondrial membrane potential.

In porcine oocyte in vitro aging models, quercetin at 10 μM restored ΔΨm and ATP content, suppressed abnormal autophagy and apoptosis, and rescued the expression of maturation-promoting factors [41]. Kim et al. [42] further demonstrated that quercetin delays postovulatory aging by preventing the downregulation of SIRT1, SIRT2, and SIRT3, preserving spindle integrity and normal mitochondrial distribution patterns. Epigallocatechin-3-gallate (EGCG), the most abundant polyphenol of green tea (*Camellia sinensis*), exhibits multifaceted mitochondrial protective effects in reproductive cells. Zhang et al. [54] demonstrated that EGCG treatment of aging oocytes improved ΔΨm, mitochondrial distribution, ROS levels, and spindle morphology in aged mouse oocytes in vitro and in vivo. The comprehensive review by Zhang et al. [34] established that EGCG modulates fertility by regulating ROS levels and influencing the expression and activity of catalase, SOD1, SOD2, and GPx. In porcine postovulatory aging models, Zhou et al. [35] reported that EGCG at 5 μM augmented active mitochondrial number, elevated ATP levels, and upregulated PINK1 expression, suggesting enhancement of mitophagy efficiency as a component of its mitochondrial quality-control mechanism. Huang et al. [43] established that EGCG supplementation during porcine IVM and IVF reduced ROS, increased glutathione content and total antioxidant capacity, and improved cleavage and blastocyst rates.

6.2 | Carotenoids and Mitochondrial Redox Protection

Carotenoids are lipid-soluble isoprenoid pigments characterized by extended conjugated polyene chains that confer exceptional free radical-quenching capacity. Their hydrophobic character enables preferential partitioning into the inner mitochondrial membrane, positioning them as singularly effective interceptors of oxidative damage at its principal cellular source [23, 44]. Astaxanthin (3,3′-dihydroxy-β-β-carotene-4,4′-dione), biosynthesized primarily by *Haematococcus pluvialis*, possesses antioxidant activity that substantially exceeds that of β-carotene and α-tocopherol by virtue of its unique molecular architecture enabling ROS scavenging across both the aqueous-membrane interfaces and the hydrophobic core of biological bilayers.

In male reproductive biology, Liu et al. [55] demonstrated that astaxanthin treatment improved sperm concentration and forward motility in oligoasthenozoospermic mice, reduced mitochondrial oxidative stress in spermatogonial stem cells, and regulated the ferroptosis pathway. Eslami et al. [33] established that astaxanthin activates the Nrf2/ARE pathway in human granulosa cells. The double-blind randomized trial of Comhaire et al. [56] documented that combined antioxidant treatment incorporating astaxanthin produced significant improvements in progressive sperm motility and clinical pregnancy rates in infertile men. In ram spermatozoa, Salman et al. [45] demonstrated that 3.5 μM astaxanthin supplementation significantly elevated ΔΨm and total antioxidant capacity while reducing ROS and lipid peroxidation markers.

Lycopene, the red acyclic carotenoid present in tomatoes and other red-pigmented fruits, constitutes the most potent singlet

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oxygen quencher among the naturally occurring carotenoids [23]. In male reproductive biology, lycopene accumulates preferentially in testicular tissue and the mitochondria-enriched midpiece of spermatozoa, where it modulates mitochondrial membrane integrity and augments catalase and GPx activities. Babaei et al. [47] demonstrated in a varicocele animal model that lycopene administration at 10 mg/kg significantly improved sperm concentration and motility, elevated total antioxidant capacity, and reduced ROS and DNA damage. The randomized double-blind placebo-controlled trial of Nouri et al. [46] in 44 infertile oligozoospermic men provided clinical corroboration: 25 mg/day lycopene supplementation for 12 weeks produced significant improvements in total sperm count and concentration relative to placebo.

6.3 | Terpenoids and Mitochondrial Signaling Modulation

Several terpenoids of medicinal plant origin have demonstrated significant modulatory effects on mitochondrial signaling pathways of direct relevance to reproductive aging. Ginsenoside Rg1 supplementation ameliorated premature ovarian insufficiency in a murine model by preserving GC mitochondrial function, attenuating apoptosis through BCL-2/BAX modulation, and activating SIRT1-dependent transcriptional programs [57]. Network pharmacological analysis by Liu et al. [58] identified multiple molecular targets through which *Panax ginseng* components address premature ovarian insufficiency, including mitochondria-associated apoptotic and oxidative stress pathway proteins. Ursolic acid and oleanolic acid activate AMPK signaling and improve mitochondrial respiration efficiency in metabolically stressed cells, providing antiapoptotic protection in GCs, effects of particular relevance to the metabolic origins of poor oocyte quality in obese and insulin-resistant patients [31].

6.4 | Alkaloids and Bioenergetic Reprogramming

Berberine, an isoquinoline alkaloid isolated principally from *Coptis chinensis* and *Berberis vulgaris*, has emerged as the most extensively characterized alkaloid in reproductive mitochondrial pharmacology. Berberine's primary molecular mechanism entails partial reversible inhibition of mitochondrial ETC Complex I, which activates AMPK through the consequent elevation of the intracellular AMP:ATP ratio [37]. AMPK activation initiates a comprehensive bioenergetic reprogramming cascade: phosphorylation and activation of PGC-1α stimulates mitochondrial biogenesis; phosphorylation of acetyl-CoA carboxylase promotes fatty acid oxidation; and inhibition of mTORC1 redirects metabolic resources toward catabolic pathways [36]. Song et al. [48] demonstrated in diabetic rats that berberine significantly improved sperm motility, morphology, and spermatogenic epithelium integrity while suppressing ROS-mediated JAK2/NFκB inflammatory signaling. Hassani-Bafrani et al. [49] demonstrated in a varicocele rat model that berberine attenuated oxidative-inflammatory testicular damage, elevated total antioxidant capacity, and reduced IL-6 and TNF-α levels.

6.5 | Coenzyme Q10 as an Isoprenoid Mitochondrial Electron Carrier

Coenzyme Q10 (CoQ10; ubiquinone) is a lipid-soluble isoprenoid that functions as an obligate electron shuttle in the mitochondrial ETC, transferring electrons from Complexes I and II to Complex III and thereby coupling OXPHOS to ATP synthesis. It additionally functions in its reduced form (ubiquinol) as a potent lipophilic antioxidant within the inner mitochondrial membrane, quenching lipid peroxyl radicals and regenerating vitamins C and E [50, 52]. The landmark study by Ben-Meir et al. [50] demonstrated that aging of the female germline is accompanied by mitochondrial dysfunction associated with decreased OXPHOS and reduced ATP, attributed to age-dependent downregulation of CoQ biosynthetic enzymes (Pdss2, Coq6); remarkably, dietary CoQ10 administration reversed the age-related decline in oocyte quality and quantity, prevented premature ovarian failure in Pdss2-deficient mice, and reduced spindle abnormalities. The prospective RCT by Xu et al. [51] in young women with diminished ovarian reserve demonstrated that 600 mg/day CoQ10 for 60 days prior to IVF significantly increased oocyte yield, fertilization rate, and high-quality embryo number relative to controls. A systematic review and meta-analysis by Lin et al. [59] encompassing six RCTs and 1529 participants confirmed that CoQ10 pretreatment is significantly correlated with elevated clinical pregnancy rate, increased oocyte retrieval, and reduced cycle cancellation rate in women with diminished ovarian reserve.

7 | Phytochemicals in Fertility Preservation and Assisted Reproductive Technology

The translation of phytochemical mitochondrial modulation from preclinical experimental systems to clinical ART applications represents an actively evolving research frontier. The mechanistic rationale is compelling: given the well-established role of mitochondrial dysfunction in oocyte developmental competence impairment, sperm functional deterioration, and embryo developmental failure, adjunctive phytochemical therapy that restores or enhances mitochondrial bioenergetic integrity could plausibly improve the efficiency and outcomes of IVF, ICSI, fertility preservation, and gamete cryopreservation procedures [38].

Table 2 summarizes the key preclinical and clinical studies examining phytochemical effects on female reproductive outcomes, while Table 3 presents the corresponding evidence for male reproductive parameters. For female fertility, the randomized triple-blind trial of Mohamadhasani et al. [39] in women with PCOS established proof-of-concept that supraphysiological resveratrol supplementation (800 mg/day) produces molecularly verifiable GC mitochondrial biogenesis enhancement that translates into significantly improved oocyte maturity and high-quality embryo formation rates. This trial is notable for its integration of mechanistic biomarker assessment in the same patient cohort, providing a level of evidence that strengthens the causal inference between mitochondrial biogenesis activation and ART outcome improvement.

Quercetin's demonstrated capacity to improve human oocyte IVM rates by 19.6% and blastocyst formation rates by 15.5% in supplemented culture systems [40] supports its potential

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TABLE 2 | Selected preclinical and clinical studies on phytochemicals and female reproductive aging or ART outcomes.

Phytochemical	Study type	Population/model	Dose/concentration	Key mitochondrial & reproductive findings	References (DOI)[a]
Resveratrol	Randomized triple-blind placebo-controlled clinical trial	Women with PCOS undergoing IVF ($n = 62$)	800 mg/day for 40 days	↑ SIRT1 and PGC-1α protein; ↑ mtDNA copy number; ↑ ATP in GCs; ↑ oocyte maturity rate; ↑ high-quality embryo rate; ↓ total oxidant status in follicular fluid	Mohamadhasani et al. [39]
Resveratrol	In vivo animal experiment	Aging ICR mice (47 weeks; $n = 40$)	Oral administration; 1–22 weeks	↑ ΔΨm and ATP content in oocytes; ↑ implantation rate; ↑ live offspring rate; ↓ abortion rate; ↑ Sirt1–7 ovarian expression	Okamoto et al. [53]
Quercetin	In vitro (murine and human oocytes)	Aged mouse oocytes + human oocytes	10–20 μM in IVM medium	↑ IVM rate by 19.6%; ↑ blastocyst formation by 15.5%; ↓ mitochondrial ROS via SIRT3/SOD2; ↑ spindle morphology; ↓ apoptosis	Cao et al. [40]
Quercetin	In vitro animal experiment	Porcine oocytes (in vitro aging model)	10 μM in maturation medium	↑ ΔΨm; ↑ ATP content; ↑ oocyte maturation rate; ↓ abnormal autophagy; ↓ apoptosis (↑ BCL2, ↓ BAX); ↑ MPF expression	Jiao et al. [41]
EGCG	In vitro and in vivo animal experiment	Aged ICR mouse oocytes (9–10 months)	10 μM in vitro; IP injection in vivo	↑ first polar body extrusion; ↑ ΔΨm; ↑ mitochondrial distribution; ↓ intracellular ROS; ↑ spindle morphology; ↑ in vivo fertilization rate	Zhang et al. [54]
EGCG	In vitro animal experiment	Porcine postovulatory aged oocytes (MII; 48 h)	5 μM in aging medium	↑ active mitochondrial count; ↑ ATP; ↑ PINK1 expression; ↓ autophagy; ↓ apoptosis; ↑ blastocyst development	Zhou et al. [35]
Coenzyme Q10	Randomized controlled trial	Young women with POR (POSEIDON group 3, age < 35; $n = 169$)	600 mg/day for 60 days pre-IVF	↑ oocytes retrieved; ↑ fertilization rate; ↑ high-quality embryos; ↓ gonadotrophin requirements; ↑ peak E2 level	Xu et al. [51]
Coenzyme Q10	In vivo animal experiment + human tissue analysis	Aged female mice + human oocytes	Dietary administration (15 weeks)	↓ OXPHOS impairment; ↑ ATP; ↑ ovarian reserve; ↓ spindle abnormalities; ↑ aneuploidy-free oocytes; ↑ live birth rate; reversed by Pdss2 KO	Ben-Meir et al. [50]

Abbreviations: ART, assisted reproductive technology; ATP, adenosine triphosphate; EGCG, epigallocatechin-3-gallate; GC, granulosa cell; IVF, in vitro fertilization; IVM, in vitro maturation; mtDNA, mitochondrial DNA; OXPHOS, oxidative phosphorylation; PCOS, polycystic ovary syndrome; POR, poor ovarian response; RCT, randomized controlled trial; ROS, reactive oxygen species; SIRT, sirtuin; ΔΨm, mitochondrial membrane potential.
[a]In-text citations follow APA (Author, year) format.

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TABLE 3 | Selected preclinical and clinical studies on phytochemicals and male reproductive parameters.

Phytochemical	Study type	Population/model	Dose/concentration	Key sperm/testicular findings	References (DOI)[a]
Astaxanthin	Double-blind randomized controlled trial	Infertile men ($n = 30$ per arm)	Not individually specified; combined antioxidant regimen	↑ progressive sperm motility; ↑ clinical pregnancy rate per cycle; ↑ fertilization rate compared with conventional treatment alone	Comhaire et al. [56]
Astaxanthin	Network pharmacology + in vivo animal experiment	Oligoasthenozoospermic mice (cyclophosphamide-induced)	10, 50, and 100 mg/kg/day (oral, 21 days)	↑ sperm concentration and forward motility; ↓ mitochondrial oxidative stress; ↑ GPX4, GSH-Px, FADS2; regulated ferroptosis/fatty acid pathway	Liu et al. [55]
Lycopene	Randomized double-blind placebo-controlled trial	Infertile men with oligozoospermia ($n = 44$)	25 mg/day for 12 weeks	↑ total sperm count; ↑ sperm concentration; ↓ seminal oxidative stress (MDA, TAC improvement); antioxidant enzyme activity modulation	Nouri et al. [46]
Lycopene	In vivo animal experiment (varicocele model)	Adult male Wistar rats with induced varicocele ($n = 45$)	4 or 10 mg/kg for 2 months	↑ sperm concentration and motility (at 10 mg/kg); ↑ TAC; ↑ CAT activity; ↓ ROS; ↓ MDA; ↓ sperm DNA damage index	Babaei et al. [47]
Berberine	In vivo animal experiment	Streptozotocin-induced diabetic male Sprague–Dawley rats	Oral administration for 4 weeks	↑ sperm motility and normal morphology; ↑ testis weight; ↑ spermatogenic epithelium integrity; ↓ ROS; ↓ JAK2/NFκB inflammatory signaling; ↑ fertility rate	Song et al. [48]
Berberine	In vivo animal experiment (varicocele model)	Mature male Wistar rats with experimental varicocele ($n = 50$)	50 or 100 mg/kg (oral, 60 days)	↑ TAC; ↓ IL-6; ↓ TNF-α; ↓ NO; ↓ MDA; ↑ spermatogenic cell density; improved testicular and epididymal histology	Hassani-Bafrani et al. [49]
Lycopene	Systematic review	Human and animal studies; multiple designs	Variable (10–30 mg/day in human studies)	↓ lipid peroxidation; ↓ sperm DNA damage; ↑ sperm count/viability; ↑ antioxidant enzyme activities (CAT, GPx); modulates mitochondrial membrane integrity	Durairajanayagam et al. [23]

Abbreviations: CAT, catalase; DNA DFI, DNA fragmentation index; GPx, glutathione peroxidase; IL-6, interleukin-6; MDA, malondialdehyde; RCT, randomized controlled trial; ROS, reactive oxygen species; SOD, superoxide dismutase; TAC, total antioxidant capacity; TNF-α, tumor necrosis factor-alpha.
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application as an IVM medium additive. EGCG supplementation of IVM and IVF media has similarly been shown to reduce intracellular ROS, improve cumulus cell expansion, and enhance subsequent developmental competence in porcine experimental systems [43], with analogous beneficial effects on aging mouse oocytes in vivo [54]. The prospective RCT of Xu et al. [51] and the meta-analysis of Lin et al. [59] collectively demonstrate that CoQ10 pretreatment increases oocyte yield, fertilization rates, high-quality embryo rates, and clinical pregnancy rates in women with diminished ovarian reserve.

In male reproductive medicine, the double-blind randomized trial by Comhaire et al. [56] established clinical-grade evidence that combined antioxidant treatment incorporating astaxanthin improves sperm progressive motility and pregnancy rates in infertile men, while the RCT of Nouri et al. [46] demonstrated that lycopene supplementation significantly increases sperm count and concentration in oligozoospermic men over a 12-week treatment course. These findings are mechanistically coherent with the established roles of these carotenoids in stabilizing sperm mitochondrial membranes, reducing mitochondrial ROS, and protecting sperm chromatin integrity [23, 47].

The preclinical evidence from Song et al. [48] and Hassani-Bafrani et al. [49] for berberine in spermatogenic protection through AMPK-mediated anti-oxidative and anti-inflammatory mechanisms provides mechanistic grounding for ongoing translational investigation of berberine in male infertility management. The application of phytochemicals to gamete cryopreservation media represents an especially promising translational direction. The freeze–thaw process imposes profound oxidative stresses upon gametes, including mitochondrial ROS generation from ETC disruption during temperature transitions and lipid peroxidation of membrane fatty acids [44].

Supplementation of sperm cryopreservation extenders with astaxanthin consistently reduces post-thaw ROS generation, preserves ΔΨm, and improves sperm motility and viability parameters across multiple species [45]. In the challenging scenarios of diminished ovarian reserve and premature ovarian insufficiency, phytochemical mitochondrial modulation offers a theoretically rational adjunctive strategy for improving the quality of the limited remaining follicular pool. Ginsenoside Rg1 supplementation has demonstrated efficacy in premature ovarian insufficiency animal models through SIRT1-dependent mitochondrial preservation [57, 58], and resveratrol has shown capacity to improve GC mitochondrial function under hypoxic and age-simulating conditions [25].

8 | Advanced Mitochondria-Targeted Delivery Systems in Reproductive Phytomedicine

Despite the growing evidence supporting phytochemicals as modulators of mitochondrial function in reproductive aging, their clinical translation remains limited by several pharmacokinetic challenges. Many bioactive compounds, including resveratrol, quercetin, EGCG, and berberine, exhibit poor aqueous solubility, rapid metabolic degradation, low oral bioavailability, and limited tissue penetration. These limitations are particularly relevant in reproductive tissues, where effective intracellular delivery into mitochondria-rich oocytes, granulosa cells, and spermatozoa is required to achieve meaningful biological effects. As a result, increasing attention has shifted toward the development of mitochondria-targeted delivery systems designed to improve intracellular accumulation, stability, and therapeutic efficacy [38, 60] (Figure 3).

Among the most extensively investigated strategies are triphenylphosphonium (TPP^+)-based delivery systems. TPP^+ is a lipophilic cation that preferentially accumulates within mitochondria due to the highly negative mitochondrial membrane potential across the inner mitochondrial membrane. Conjugation of phytochemicals or antioxidant molecules to TPP^+ substantially enhances mitochondrial localization and improves ROS scavenging directly at the major site of oxidative stress generation. Experimental studies have demonstrated that TPP^+-linked antioxidants improve mitochondrial respiration, preserve ATP synthesis, stabilize ΔΨm, and reduce mitochondrial lipid peroxidation more effectively than unconjugated compounds [61, 62]. In the context of reproductive biology, this strategy may be particularly relevant because oocytes and spermatozoa possess exceptionally high mitochondrial density and are highly susceptible to oxidative injury during aging and assisted reproductive procedures.

Nanotechnology-based delivery systems are also emerging as promising tools for mitochondrial modulation in reproductive medicine. Polymeric nanoparticles, solid lipid nanoparticles, dendrimers, and mesoporous silica nanoparticles can improve phytochemical stability, prolong circulation time, and enhance intracellular uptake [63]. Functionalization of these nanoparticles with mitochondrial-targeting ligands such as TPP^+ or mitochondria-penetrating peptides allows selective accumulation within dysfunctional mitochondria, thereby increasing therapeutic efficiency while reducing systemic exposure [64]. In aging ovaries, where mitochondrial dysfunction, hypoxia, and oxidative stress coexist within the follicular microenvironment, targeted nanocarriers may improve local bioavailability of mitochondria-protective compounds and enhance follicular bioenergetics.

Liposomal formulations represent another attractive platform because of their biocompatibility and ability to encapsulate both hydrophilic and lipophilic molecules. Liposomes protect phytochemicals from premature degradation and facilitate controlled intracellular delivery. Mitochondria-targeted liposomes modified with cationic phospholipids or TPP^+ derivatives have shown enhanced mitochondrial uptake and improved intracellular antioxidant activity in several experimental systems [65]. In reproductive medicine, liposomal delivery systems may be particularly useful during IVF and oocyte in vitro maturation, where supplementation can be applied directly to culture media to improve mitochondrial stability and reduce oxidative damage during embryo development and cryopreservation.

Exosome-based delivery systems have also attracted growing attention in reproductive regenerative medicine. Exosomes are naturally derived extracellular vesicles involved in intercellular communication and molecular transport. Their endogenous origin provides advantages including low immunogenicity, high biological compatibility, and intrinsic tissue-targeting capability [66]. In experimental models of ovarian insufficiency and

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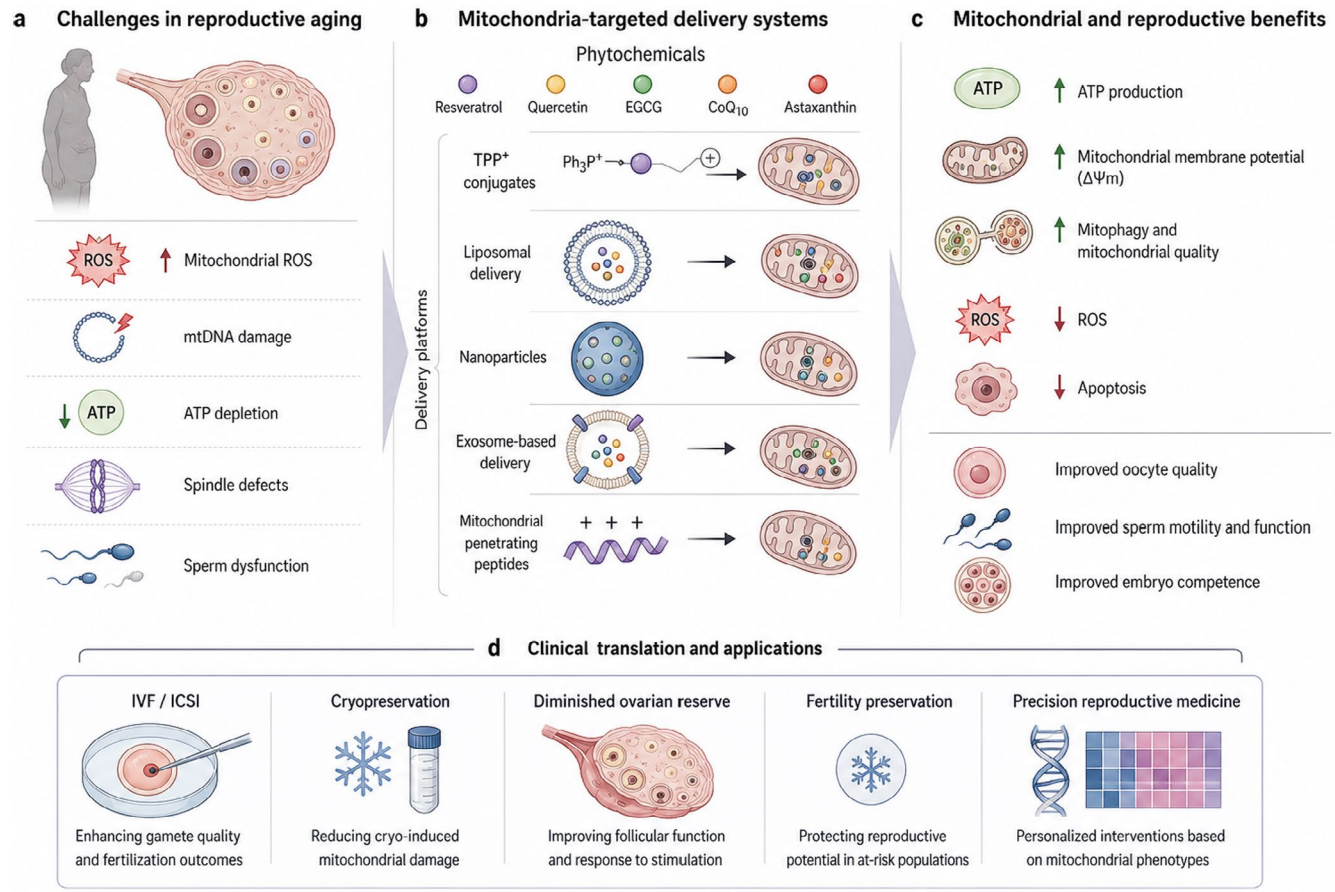


FIGURE 3 | Mitochondria-targeted delivery systems for phytochemical modulation in reproductive aging. The figure summarizes the relationship between reproductive aging–associated mitochondrial dysfunction, advanced mitochondrial delivery platforms, and their potential translational applications in reproductive medicine. Aging-related reproductive decline is characterized by increased mitochondrial ROS generation, mtDNA damage, ATP depletion, spindle abnormalities, and impaired sperm function. Emerging delivery systems, including TPP^+ conjugates, liposomes, nanoparticles, exosome-based carriers, and mitochondrial-penetrating peptides, are designed to improve mitochondrial targeting and intracellular delivery of phytochemicals such as resveratrol, quercetin, EGCG, CoQ10, and astaxanthin. Enhanced mitochondrial delivery contributes to restoration of mitochondrial homeostasis through increased ATP production, preservation of mitochondrial membrane potential (ΔΨm), improved mitophagy, and reduced oxidative stress and apoptosis. These effects may ultimately improve oocyte quality, sperm motility, embryo competence, and clinical outcomes in IVF/ICSI, cryopreservation, fertility preservation, and precision reproductive medicine.

testicular injury, mesenchymal stem cell-derived exosomes have been shown to restore mitochondrial homeostasis, reduce oxidative stress, and improve reproductive tissue repair [67, 68]. Engineering exosomes to carry mitochondria-modulating phytochemicals or mitochondrial regulatory RNAs may therefore provide a biologically adaptive strategy for improving reproductive bioenergetics and preserving gamete quality during aging.

Another emerging concept involves ovarian-targeted nanocarriers designed to selectively accumulate within the follicular compartment. Surface modification with ligands recognizing follicle-stimulating hormone receptor (FSHR) or granulosa-cell–associated proteins may improve tissue specificity and reduce off-target distribution [69]. Such approaches could be particularly relevant for women with diminished ovarian reserve or poor ovarian response, where preservation of the remaining follicular pool is critically important. Similarly, targeted delivery into Sertoli cells and spermatogonial niches may offer therapeutic opportunities in age-related male infertility and oxidative testicular injury.

Although these technologies remain largely experimental, they represent an important shift from conventional antioxidant supplementation toward organelle-specific therapeutic modulation. Future progress will depend not only on improving targeting efficiency and pharmacokinetic performance, but also on addressing challenges related to long-term biosafety, reproductive toxicology, biodegradability, manufacturing reproducibility, and regulatory standardization. Integration of mitochondrial phenotyping, multi-omics profiling, and advanced delivery engineering may ultimately facilitate the development of precision reproductive therapeutics tailored to individual mitochondrial dysfunction patterns.

9 | Translational Challenges and Clinical Considerations

Notwithstanding the substantial and mechanistically coherent body of preclinical and early clinical evidence reviewed herein, the translation of phytochemical mitochondrial

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modulation strategies into established, evidence-based clinical fertility interventions confronts a constellation of methodological, pharmacological, and regulatory challenges. The most consequential gap is the paucity of rigorously designed, adequately powered, prospectively registered randomized controlled trials that evaluate phytochemical interventions against placebo with prespecified fertility outcome endpoints in well-defined patient populations. The majority of available evidence derives from preclinical experimental models, often employing supraphysiological compound concentrations with uncertain human pharmacokinetic relevance [31, 38]. Prioritized future clinical investigation should include large, multicenter RCTs with concurrent mechanistic biomarker assessment incorporating mtDNA copy number quantification, ΔΨm measurement in GCs, and intracellular ATP content, alongside primary clinical endpoints of live birth and cumulative ART success rates.

Bioavailability limitations represent a fundamental pharmacological constraint for the most mechanistically promising polyphenolic phytochemicals. Resveratrol, quercetin, EGCG, and structurally related compounds are characterized by poor and highly variable oral bioavailability, attributable to extensive phase II conjugation in the intestinal mucosa and liver, rapid plasma clearance, and poor aqueous solubility [3]. Strategies to overcome this limitation include nanotechnology-enabled delivery systems (polymeric nanoparticles, liposomes, solid lipid nanoparticles), micellar formulations, cyclodextrin inclusion complexes, and coadministration with bioavailability enhancers such as piperine. Variability of herbal preparations, potential endocrine interactions—including the phytoestrogenic or antiestrogenic activity of certain polyphenols and ginsenosides—and the regulatory gap between nutraceutical and pharmaceutical standards represent additional translational impediments requiring systematic investigation [3, 57] (Figure 4).

9.1 | Safety Considerations and Potential Adverse Effects

The preclinical profile of these compounds should not be mistaken for an absence of risk. Because the intended recipients are frequently women attempting conception, women already pregnant, or men in couples pursuing treatment, the reproductive

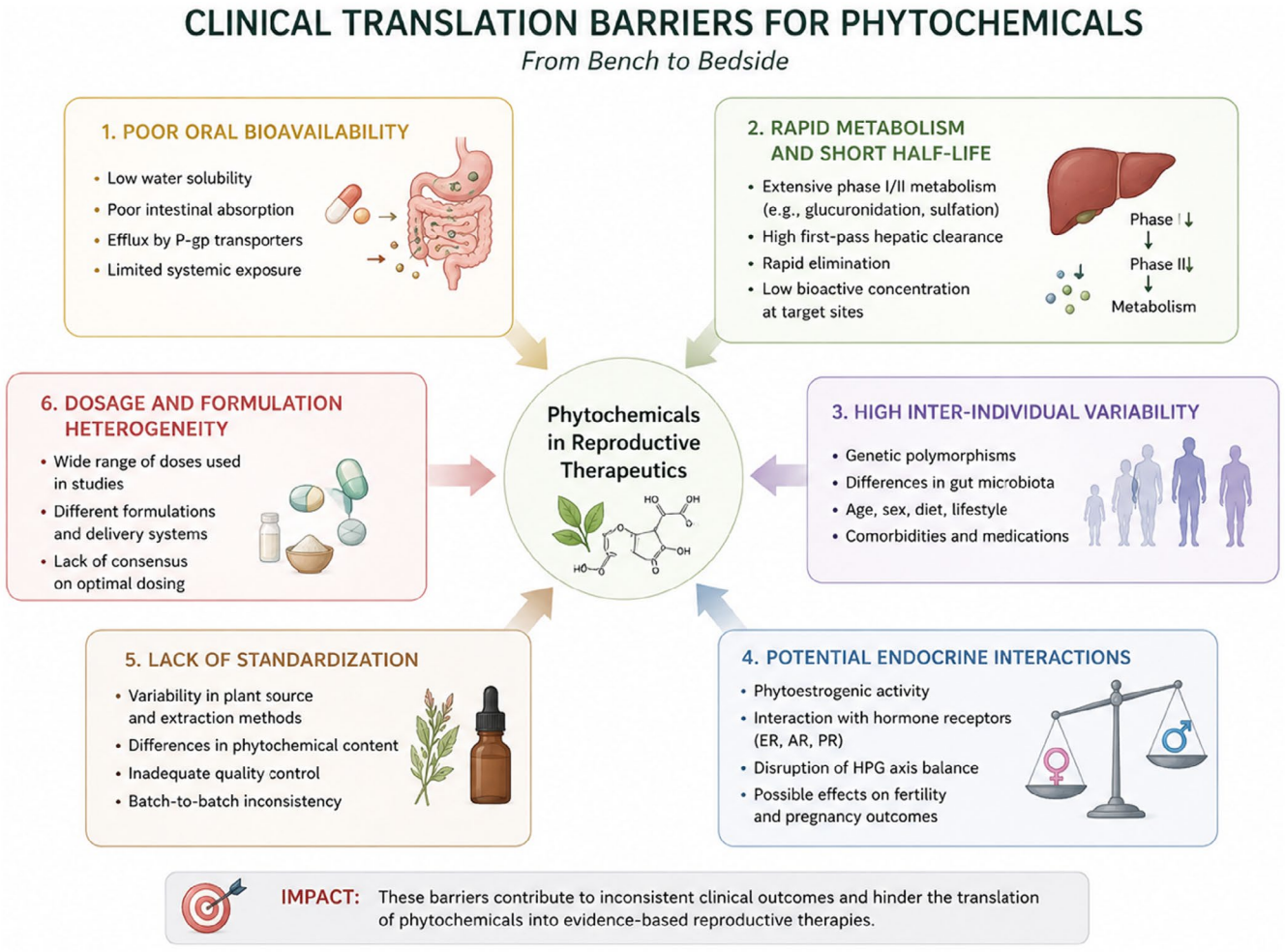


FIGURE 4 | Clinical translation barriers limiting the therapeutic application of phytochemicals in reproductive medicine. The figure summarizes the major challenges affecting the clinical translation of mitochondria-targeted phytochemicals for reproductive aging and fertility preservation, including poor oral bioavailability, rapid metabolism and clearance, interindividual variability, potential endocrine interactions, lack of formulation standardization, and dosage heterogeneity. Collectively, these factors contribute to inconsistent pharmacokinetics, variable therapeutic responses, and limited clinical reproducibility despite promising preclinical evidence.

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and developmental safety of each phytochemical requires explicit evaluation rather than an assumption of harmlessness based on plant origin. The available data reveal compound-specific toxicities, dose ceilings, and drug interactions that bear directly on clinical use.

Resveratrol is well tolerated in short-term human studies at oral doses up to approximately 1 g/day, above which gastrointestinal symptoms such as nausea and loose stools become common [70]. Its reproductive effects are not uniformly favorable. Resveratrol shows both estrogenic and antiestrogenic activity and has been reported to impair endometrial decidualization and implantation in experimental models; a clinical analysis of IVF–embryo transfer cycles linked preconception resveratrol intake to lower clinical pregnancy and higher miscarriage rates, in contrast to the improved oocyte maturity and embryo quality reported in the PCOS trial of Mohamadhasani et al. and Ochiai and Kuroda [39, 70]. Timing and patient population therefore determine whether resveratrol helps or harms, which argues against indiscriminate periconceptional supplementation.

Green tea catechins carry a defined hepatotoxic ceiling. The European Food Safety Authority concluded that EGCG at or above 800 mg/day taken as a supplement produces statistically significant elevations of serum transaminases and may indicate early liver injury, whereas catechin intake from brewed tea is generally safe [71]. A therapeutic EGCG regimen should remain below this threshold and include monitoring of liver enzymes.

Quercetin shows no evidence of toxicity at oral doses up to 1 g/day for as long as 12 weeks, but long-term high-dose data are lacking, it can act as a prooxidant under some conditions, and it may worsen injury in an already compromised kidney; dose-limiting nephrotoxicity has been observed only with high intravenous exposure [72]. Quercetin also inhibits several cytochrome P450 enzymes and efflux transporters, creating potential interactions with coadministered drugs.

Berberine warrants the greatest caution in a reproductive context and is contraindicated in pregnancy and lactation. It displaces bilirubin from albumin, roughly tenfold more potently than phenylbutazone in vitro, and can raise unbound bilirubin, a mechanism linked to kernicterus risk in the neonate [73]. It inhibits CYP3A4, CYP2D6, and P-glycoprotein, producing clinically meaningful interactions with statins, cyclosporine, and other substrates, and its glucose-lowering action can compound that of antidiabetic agents. Its spermatogenic benefits therefore apply to male use rather than to periconceptional use in women.

Ginsenoside-containing preparations may exhibit estrogenic activity and can reduce the anticoagulant effect of warfarin, as shown in a randomized trial of American ginseng [74]. Wide product-to-product variation in ginsenoside content adds further uncertainty.

Astaxanthin, lycopene, and coenzyme Q10 present the most reassuring profiles. Coenzyme Q10 has been tolerated at doses up to 1200–1800 mg/day with only mild gastrointestinal complaints, though it may attenuate warfarin activity [75]; astaxanthin and lycopene are generally well tolerated, with high lycopene intake occasionally producing benign skin discoloration.

Two problems affect the whole class. Herbal and nutraceutical preparations vary widely in active-compound content and purity because they are not held to pharmaceutical manufacturing standards, so a labeled dose need not reflect the delivered dose. Several of these compounds are also pharmacologically active at the endocrine and metabolic level, which means that supplement status does not exempt them from the interaction and reproductive-toxicity assessment expected of any agent used around conception. Systematic dose-ranging and reproductive-safety studies are a prerequisite for clinical adoption.

10 | Future Directions: Toward Mitochondrial Precision Medicine in Reproductive Health

The growing convergence of mitochondrial biology, reproductive medicine, systems pharmacology, and nanotechnology is reshaping current perspectives on fertility preservation and reproductive aging. Increasing evidence suggests that future therapeutic strategies will move beyond conventional antioxidant supplementation toward more selective approaches aimed at restoring mitochondrial integrity, bioenergetic stability, and organelle quality control within reproductive tissues. In this context, mitochondria-targeted phytochemical delivery systems represent a particularly promising direction for translational development.

Mitochondria-targeted derivatives, formed by conjugating phytochemicals to TPP^+ cations or mitochondria-penetrating peptides, are one such strategy and are detailed in Section 8; they concentrate the parent compound within mitochondria and reduce off-target exposure [31, 38].

Another important future direction involves the development of personalized reproductive interventions guided by mitochondrial phenotyping. Quantitative assessment of mtDNA copy number, mutational burden, $\Delta\Psi m$, oxidative stress signatures, and cellular bioenergetic capacity in granulosa cells or oocytes may eventually allow identification of patients most likely to benefit from mitochondria-targeted therapies [14, 18]. Such stratification approaches may be particularly relevant in women with diminished ovarian reserve, recurrent implantation failure, poor ovarian response, or age-related infertility, where mitochondrial dysfunction appears to play a central pathogenic role.

In practical terms, several of these parameters are already measurable in the embryology laboratory from material that would otherwise be discarded. The mitochondrial DNA copy number of cumulus and mural granulosa cells, quantified by real-time PCR against a nuclear reference gene, correlates with oocyte competence and embryo quality and can be obtained noninvasively from the cumulus mass removed before ICSI [76]. Bioenergetic capacity of the same cells can be profiled by extracellular-flux respirometry to yield basal and maximal oxygen consumption and spare respiratory capacity;

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ΔΨm can be estimated cytofluorometrically with JC-1 in granulosa cells or spermatozoa; and oxidative-stress load can be indexed through follicular-fluid or seminal markers of lipid peroxidation. Because absolute mtDNA thresholds differ between patients and laboratories, these readings are most informative as a within-patient profile rather than a universal cut-off [76].

A phenotype-guided pathway would translate these readings into specific choices. Granulosa cells showing low mtDNA copy number and reduced spare respiratory capacity, signaling a biogenesis deficit, would direct selection toward biogenesis-promoting agents such as resveratrol or coenzyme Q10, the latter supported by randomized evidence in diminished ovarian reserve [51, 59]. A profile dominated by high oxidative-stress markers with relatively preserved copy number would instead favor redox-protective compounds such as astaxanthin or lycopene. Men with low sperm ΔΨm and elevated DNA fragmentation would be candidates for carotenoid or coenzyme Q10 supplementation. Repeat sampling in a later cycle would then test whether the intervention shifted the targeted parameter, linking measurement to treatment. Applied this way, mitochondrial phenotyping converts phytochemical supplementation from an undifferentiated adjuvant into a stratified intervention matched to the dominant defect in a given patient.

Combination regimens that pair a biogenesis promoter with a redox-protective agent, discussed in Section 5, extend this phenotype-guided logic and merit direct testing in reproductive aging.

The gut microbiome has additionally emerged as a potentially important determinant of phytochemical responsiveness. Interindividual differences in microbial metabolism may substantially influence the absorption, biotransformation, and biological activity of resveratrol, quercetin, EGCG, and other polyphenolic compounds, potentially contributing to the variability observed across clinical studies [3]. Future studies integrating microbiome profiling with mitochondrial biomarkers may therefore improve understanding of treatment responsiveness and support the development of more individualized phytochemical interventions.

Emerging regenerative approaches may further broaden the scope of mitochondrial therapeutics in reproductive medicine. Extracellular vesicle–based delivery systems, stem cell-derived exosomes, and mitochondrial transfer strategies are increasingly being investigated for their potential to restore mitochondrial function and improve ovarian or testicular microenvironmental homeostasis. Although these approaches remain largely experimental, they highlight the growing shift toward organelle-directed reproductive therapies aimed not only at delaying reproductive decline but also at restoring cellular bioenergetic competence.

These advances suggest that the future of reproductive therapeutics may increasingly depend on precise modulation of mitochondrial function rather than solely on hormonal manipulation or symptomatic intervention. As understanding of reproductive bioenergetics continues to evolve, mitochondria-targeted phytochemical strategies may become an important component of next-generation fertility preservation and reproductive rejuvenation approaches.

11 | Conclusions

This review highlights the central role of mitochondrial dysfunction in the biology of reproductive aging and supports the view that declining reproductive capacity is closely linked to progressive impairment of cellular bioenergetics. In both female and male reproductive tissues, excessive mitochondrial ROS generation, mtDNA instability, impaired mitochondrial quality control, and reduced oxidative phosphorylation collectively contribute to the functional decline of oocytes, granulosa cells, and spermatozoa. These mitochondrial alterations are increasingly recognized not simply as secondary features of aging, but as active drivers of reproductive senescence and reduced fertility potential.

Within this framework, phytochemicals have emerged as biologically plausible modulators of mitochondrial homeostasis. Rather than acting through a single pathway, many phytochemicals exert coordinated effects across interconnected mitochondrial networks involved in oxidative stress regulation, mitochondrial biogenesis, mitophagy, metabolic adaptation, and apoptotic signaling. This systems-level activity may be particularly relevant in reproductive aging, where mitochondrial dysfunction develops through multiple overlapping mechanisms. Experimental and early clinical evidence collectively suggests that mitochondria-targeted phytochemical interventions may improve reproductive bioenergetics, preserve gamete quality, and support embryo developmental competence, particularly in the setting of age-related infertility and assisted reproductive technologies.

At the same time, several important translational barriers remain unresolved. The current evidence base is still dominated by preclinical studies, while large, well-designed randomized clinical trials remain limited. Additional challenges include poor oral bioavailability of many phytochemicals, variability in formulation quality, dosage heterogeneity, and incomplete understanding of long-term endocrine and reproductive effects. Addressing these limitations will be essential before mitochondria-targeted phytochemicals can be more fully integrated into routine clinical reproductive practice.

Recent advances in systems pharmacology, mitochondrial phenotyping, targeted nanodelivery systems, and reproductive bioenergetics are nevertheless reshaping the therapeutic landscape of fertility preservation. The growing ability to selectively modulate mitochondrial function within reproductive tissues may ultimately allow more individualized approaches for patients with diminished ovarian reserve, poor ovarian response, recurrent implantation failure, or age-related infertility. In parallel, emerging strategies involving mitochondrial-targeted delivery systems, extracellular vesicles, and bioenergetic restoration approaches continue to expand the scope of reproductive therapeutics beyond conventional hormonal intervention.

As understanding of reproductive mitochondrial biology advances, phytochemical-based mitochondrial modulation may become a component of precision reproductive medicine that

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both delays reproductive decline and helps restore cellular resilience during aging.

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The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. E. Esencan, B. Simsek, and E. Seli, "Analysis of Female Demographics in the United States: Life Expectancy, Education, Employment, Family Building Decisions, and Fertility Service Utilization," *Current Opinion in Obstetrics and Gynecology* 33 (2021): 170–177, https://doi.org/10.1097/GCO.0000000000000704.

2. J. Cao, W. Xu, Y. Liu, et al., "Trends in Maternal Age and the Relationship Between Advanced Age and Adverse Pregnancy Outcomes: A Population-Based Register Study in Wuhan, China, 2010–2017," *Public Health* 206 (2022): 8–14, https://doi.org/10.1016/j.puhe.2022.02.015.

3. J. Balasch, "Ageing and Infertility: An Overview," *Gynecological Endocrinology* 26, no. 12 (2010): 855–860, https://doi.org/10.3109/09513590.2010.501889.

4. P. J. Smock and C. R. Schwartz, "The Demography of Families: A Review of Patterns and Change," *Journal of Marriage and Family* 82 (2020): 9–34, https://doi.org/10.1111/jomf.12612.

5. F. J. Broekmans, M. R. Soules, and B. C. Fauser, "Ovarian Aging: Mechanisms and Clinical Consequences," *Endocrine Reviews* 30, no. 5 (2009): 465–493, https://doi.org/10.1210/er.2009-0006.

6. C. Tatone, F. Amicarelli, M. C. Carbone, et al., "Cellular and Molecular Aspects of Ovarian Follicle Ageing," *Human Reproduction Update* 14, no. 2 (2008): 131–142, https://doi.org/10.1093/humupd/dmm048.

7. E. Babayev and E. Seli, "Oocyte Mitochondrial Function and Reproduction," *Current Opinion in Obstetrics and Gynecology* 27, no. 3 (2015): 175–181, https://doi.org/10.1097/GCO.0000000000000164.

8. E. Seli, "Mitochondrial Dysfunction and Ovarian Aging," *Endocrinology* 161, no. 2 (2020): bqaa001, https://doi.org/10.1210/endocr/bqaa001.

9. S. Pickles, P. Vigié, S. Bhatt, and R. J. Youle, "Mitophagy and Quality Control Mechanisms in Mitochondrial Maintenance," *Current Biology* 28, no. 4 (2018): R170–R185, https://doi.org/10.1016/j.cub.2018.01.004.

10. R. Ventura-Clapier, A. Garnier, and V. Veksler, "Transcriptional Control of Mitochondrial Biogenesis: The Central Role of PGC-1alpha," *Cardiovascular Research* 79, no. 2 (2008): 208–217, https://doi.org/10.1093/cvr/cvn098.

11. A. M. Firouzabadi, M. E. Rezvani, F. Fesahat, M. Dandekhani, M. Khazaei, and M. Hasanzadeh-Taheri, "The Impact of Mitochondrial Impairments on Sperm Function and Male Fertility: A Systematic Review," *Reproductive Biology and Endocrinology* 22 (2024): 88, https://doi.org/10.1186/s12958-024-01252-4.

12. T. Wang, P. Xu, J. Yuan, et al., "Mitochondrial Dysfunction in Oocytes: Implications for Fertility and Ageing," *Journal of Ovarian Research* 18, no. 1 (2025): 186, https://doi.org/10.1186/s13048-025-01764-6.

13. F. P. Xu, H. X. Zhang, Z. H. Liu, et al., "Mechanisms of Mitochondrial Dysfunction in Ovarian Aging and Potential Interventions," *Frontiers in Endocrinology* 15 (2024): 1361289, https://doi.org/10.3389/fendo.2024.1361289.

14. T. A. Santos, S. El Shourbagy, and J. C. St John, "Mitochondrial Content Reflects Oocyte Variability and Fertilization Outcome," *Fertility and Sterility* 85, no. 3 (2006): 584–591, https://doi.org/10.1016/j.fertnstert.2005.09.017.

15. J. Van Blerkom, "Mitochondria in Early Mammalian Development," *Seminars in Cell & Developmental Biology* 20, no. 3 (2009): 354–364, https://doi.org/10.1016/j.semcdb.2008.12.005.

16. R. Pasquariello, A. F. Ermisch, E. Silva, et al., "Alterations in Oocyte Mitochondrial Number and Function Are Related to Spindle Defects and Occur With Maternal Aging in Mice and Humans," *Biology of Reproduction* 100, no. 4 (2019): 971–981, https://doi.org/10.1093/biolre/ioy248.

17. Z. Zhu, J. Xu, C. Wu, J. Guo, and X. Zhao, "Effects of Reactive Oxygen Species and Mitochondrial Dysfunction on Reproductive Aging," *Frontiers in Cell and Developmental Biology* 12 (2024): 1347286, https://doi.org/10.3389/fcell.2024.1347286.

18. S. Long, Y. Zheng, X. Deng, et al., "Maintaining Mitochondrial DNA Copy Number Mitigates ROS-Induced Oocyte Decline and Female Reproductive Aging," *Communications Biology* 7, no. 1 (2024): 1229, https://doi.org/10.1038/s42003-024-06904-4.

19. A. E. Peters, S. J. Caban, E. A. McLaughlin, et al., "The Impact of Aging on Macroautophagy in the Pre-Ovulatory Mouse Oocyte," *Frontiers in Cell and Developmental Biology* 9 (2021): 691826, https://doi.org/10.3389/fcell.2021.691826.

20. M. B. R. Alves, E. C. C. Celeghini, and C. Belleannée, "From Sperm Motility to Sperm-Borne microRNA Signatures: New Approaches to Predict Male Fertility Potential," *Frontiers in Cell and Developmental Biology* 8 (2020): 791, https://doi.org/10.3389/fcell.2020.00791.

21. D. Durairajanayagam, D. Singh, A. Agarwal, and R. Henkel, "Causes and Consequences of Sperm Mitochondrial Dysfunction," *Andrologia* 53, no. 1 (2021): e13666, https://doi.org/10.1111/andr.12953.

22. A. Agarwal, A. Aponte-Mellado, B. J. Premkumar, A. Shaman, and S. Gupta, "The Effects of Oxidative Stress on Female Reproduction: A Review," *Reproductive Biology and Endocrinology* 10 (2012): 49, https://doi.org/10.1186/1477-7827-10-49.

23. D. Durairajanayagam, A. Agarwal, C. Ong, and P. Prashast, "Lycopene and Its Antioxidant Role in the Prevention of Male Infertility—A Review," *Reproductive Biomedicine Online* 28, no. 6 (2014): 641–648, https://doi.org/10.1016/j.rbmo.2014.02.007.

24. A. Nishigaki, H. Tsubokura, T. Tsuzuki-Nakao, and H. Okada, "Hypoxia: Role of SIRT1 and the Protective Effect of Resveratrol in Ovarian

14470578, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rmb2.70086, Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Function," *Reproductive Medicine and Biology* 21, no. 1 (2022): e12428, https://doi.org/10.1002/rmb2.12428.

25. A. Nishigaki, H. Okada, R. Okamoto, et al., "Resveratrol Protects Mitochondrial Quantity by Activating SIRT1/PGC-1α Expression During Ovarian Hypoxia," *Reproductive Medicine and Biology* 19, no. 2 (2020): 152–159, https://doi.org/10.1002/rmb2.12323.

26. B. Halim, H. P. Lubis, I. Adenin, J. Angellee, and R. S. Samoedra, "Meiotic Spindle View Improves the Outcome of IVF in Poor Responders: A Retrospective Analytical Study From an Indonesian IVF Center," *JBRA Assisted Reproduction* 28, no. 1 (2024): 39–46, https://doi.org/10.5935/1518-0557.20240006.

27. B. Halim, H. P. Lubis, D. Novia, and M. Thaharuddin, "Does Oval Oocyte Have an Impact on Embryo Development in In Vitro Fertilization?," *JBRA Assisted Reproduction* 21, no. 1 (2017): 15–18, https://doi.org/10.5935/1518-0557.20170005.

28. Z. Liao, Y. Ye, Y. Chen, and S. Wang, "Quercetin Alleviates OHSS-Induced Granulosa Cell Apoptosis by Regulation of Oxidative Stress and Mitochondrial Apoptosis Pathway," *Frontiers in Molecular Biosciences* 9 (2022): 1019140, https://doi.org/10.3389/fmolb.2022.1019140.

29. S. Vertika, K. K. Singh, and S. Rajender, "Mitochondria, Spermatogenesis, and Male Infertility—An Update," *Mitochondrion* 54 (2020): 26–40, https://doi.org/10.1016/j.mito.2020.06.003.

30. S. H. Park, "Role of Phytochemicals in Treatment of Aging and Cancer: Focus on Mechanism of FOXO3 Activation," *Antioxidants* 13, no. 9 (2024): 1099, https://doi.org/10.3390/antiox13091099.

31. Y. Bentov and R. F. Casper, "The Aging Oocyte—Can Mitochondrial Function Be Improved?," *Fertility and Sterility* 99, no. 1 (2013): 18–22, https://doi.org/10.1016/j.fertnstert.2012.11.031.

32. M. T. Borra, B. C. Smith, and J. M. Denu, "Mechanism of Human SIRT1 Activation by Resveratrol," *Journal of Biological Chemistry* 280, no. 17 (2005): 17187–17195, https://doi.org/10.1074/jbc.M501250200.

33. M. Eslami, S. Esfandyari, M. Aghahosseini, et al., "Astaxanthin Protects Human Granulosa Cells Against Oxidative Stress Through Activation of NRF2/ARE Pathway and Its Downstream Phase II Enzymes," *Cell Journal* 23, no. 3 (2021): 319–328, https://doi.org/10.22074/cellj.2021.7222.

34. Y. Zhang, H. Lin, C. Liu, J. Huang, and Z. Liu, "A Review for Physiological Activities of EGCG and the Role in Improving Fertility in Humans/Mammals," *Biomedicine & Pharmacotherapy* 127 (2020): 110186, https://doi.org/10.1016/j.biopha.2020.110186.

35. D. Zhou, M. H. Sun, W. J. Jiang, et al., "Epigallocatechin-3-Gallate Protects Porcine Oocytes Against Post-Ovulatory Aging Through Inhibition of Oxidative Stress," *Aging (Albany NY)* 14, no. 21 (2022): 8633–8644, https://doi.org/10.18632/aging.204368.

36. C. K. Keerthana, T. P. Rayginia, S. C. Shifana, et al., "The Role of AMPK in Cancer Metabolism and Its Impact on the Immunomodulation of the Tumor Microenvironment," *Frontiers in Immunology* 14 (2023): 1114582, https://doi.org/10.3389/fimmu.2023.1114582.

37. N. Turner, J. Y. Li, A. Gosby, et al., "Berberine and Its More Biologically Available Derivative, Dihydroberberine, Inhibit Mitochondrial Respiratory Complex I: A Mechanism for the Action of Berberine to Activate AMP-Activated Protein Kinase and Improve Insulin Action," *Diabetes* 57, no. 5 (2008): 1414–1418, https://doi.org/10.2337/db07-1552.

38. R. M. Yildirim and E. Seli, "Mitochondria as Therapeutic Targets in Assisted Reproduction," *Human Reproduction* 39, no. 10 (2024): 2147–2159, https://doi.org/10.1093/humrep/deae196.

39. M. Mohamadhasani, S. Jahanfar, M. A. Khalili, et al., "Resveratrol Ameliorates Mitochondrial Biogenesis and Reproductive Outcomes in Women With Polycystic Ovary Syndrome Undergoing Assisted Reproduction: A Randomized, Triple-Blind, Placebo-Controlled Clinical Trial," *Journal of Ovarian Research* 17, no. 1 (2024): 134, https://doi.org/10.1186/s13048-024-01470-9.

40. Y. Cao, H. Zhao, Z. Wang, et al., "Quercetin Promotes In Vitro Maturation of Oocytes From Humans and Aged Mice," *Cell Death & Disease* 11, no. 11 (2020): 965, https://doi.org/10.1038/s41419-020-03183-5.

41. Y. Jiao, Y. Wang, T. Jiang, et al., "Quercetin Protects Porcine Oocytes From In Vitro Aging by Reducing Oxidative Stress and Maintaining the Mitochondrial Functions," *Frontiers in Cell and Developmental Biology* 10 (2022): 915898, https://doi.org/10.3389/fcell.2022.915898.

42. J. Kim, W. Jang, Y. H. Cho, H. Y. Sik, and N. H. Kim, "Quercetin Delays Postovulatory Aging of Mouse Oocytes by Regulating SIRT Expression and MPF Activity," *Oncotarget* 8, no. 30 (2017): 50227–50240, https://doi.org/10.18632/oncotarget.16219.

43. K. Huang, C. Li, F. Gao, et al., "Epigallocatechin-3-Gallate Promotes the In Vitro Maturation and Embryo Development Following IVF of Porcine Oocytes," *Drug Design, Development and Therapy* 15 (2021): 1013–1020, https://doi.org/10.2147/DDDT.S295936.

44. T. Ghantabpour, S. Parvaneh, H. Parsaie, T. Ramzanian Gilani, M. S. Tabaei, and F. Amidi, "Overview of the Effects of Astaxanthin on the Male Reproductive System Focusing on Mechanism Underlying Its Action," *European Journal of Pharmacology* 1005 (2025): 178033, https://doi.org/10.1016/j.ejphar.2025.178033.

45. R. Salman, M. I. Naseer, S. Ayaz, et al., "Astaxanthin Improved the Quality of Hu Ram Semen by Increasing the Antioxidant Capacity and Mitochondrial Potential and Mitigating Free Radicals-Induced Oxidative Damage," *Animals* 14, no. 2 (2024): 319, https://doi.org/10.3390/ani14020319.

46. M. Nouri, R. Amani, M. Nasr-Esfahani, and M. J. Tarrahi, "The Effects of Lycopene Supplement on the Spermatogram and Seminal Oxidative Stress in Infertile Men: A Randomized, Double-Blind, Placebo-Controlled Clinical Trial," *Phytotherapy Research* 33, no. 12 (2019): 3203–3211, https://doi.org/10.1002/ptr.6493.

47. A. Babaei, R. Asadpour, K. Mansouri, A. Sabrivand, and S. Kazemi-Darabadi, "Lycopene Protects Sperm From Oxidative Stress in the Experimental Varicocele Model," *Food Science & Nutrition* 9, no. 12 (2021): 6806–6817, https://doi.org/10.1002/fsn3.2632.

48. X. Song, X. Bao, H. Bai, et al., "Protective Effect of Berberine on Reproductive Function and Spermatogenesis in Diabetic Rats via Inhibition of ROS/JAK2/NFκB Pathway," *Andrology* 8, no. 5 (2020): 1379–1389, https://doi.org/10.1111/andr.12764.

49. H. Hassani-Bafrani, H. Najaran, M. Razi, and H. Rashtbari, "Berberine Ameliorates Experimental Varicocele-Induced Damages at Testis and Sperm Levels: Evidences for Oxidative Stress and Inflammation," *Andrologia* 51, no. 2 (2019): e13179, https://doi.org/10.1111/and.13179.

50. A. Ben-Meir, E. Burstein, A. Borrego-Alvarez, et al., "Coenzyme Q10 Restores Oocyte Mitochondrial Function and Fertility During Reproductive Aging," *Aging Cell* 14, no. 5 (2015): 887–895, https://doi.org/10.1111/acel.12368.

51. Y. Xu, V. Nisenblat, C. Lu, et al., "Pretreatment With Coenzyme Q10 Improves Ovarian Response and Embryo Quality in Low-Prognosis Young Women With Decreased Ovarian Reserve: A Randomized Controlled Trial," *Reproductive Biology and Endocrinology* 16, no. 1 (2018): 29, https://doi.org/10.1186/s12958-018-0343-0.

52. C. Rodríguez-Varela and E. Labarta, "Does Coenzyme Q10 Supplementation Improve Human Oocyte Quality?," *International Journal of Molecular Sciences* 22, no. 17 (2021): 9541, https://doi.org/10.3390/ijms22179541.

53. N. Okamoto, Y. Sato, Y. Kawagoe, T. Shimizu, and K. Kawamura, "Short-Term Resveratrol Treatment Restored the Quality of Oocytes in Aging Mice," *Aging (Albany NY)* 14, no. 13 (2022): 5628–5643, https://doi.org/10.18632/aging.204157.

54. H. Zhang, W. Su, R. Zhao, et al., "Epigallocatechin-3-Gallate Improves the Quality of Maternally Aged Oocytes," *Cell Proliferation* 57, no. 4 (2024): e13575, https://doi.org/10.1111/cpr.13575.

14470578, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rmb2.70086, Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

55. H. Liu, X. Chen, X. Feng, et al., "Astaxanthin Promotes Spermatogenesis in Mice by Reducing Mitochondrial Oxidative Stress Damage and Regulating Fatty Acid Metabolism and Ferroptosis," *Drug Design, Development and Therapy* 19 (2025): 7777–7794, https://doi.org/10.2147/DDDT.S522738.

56. F. H. Comhaire, Y. El Garem, A. Mahmoud, F. Eertmans, and F. Schoonjans, "Combined Conventional/Antioxidant Astaxanthin Treatment for Male Infertility: A Double Blind, Randomized Trial," *Asian Journal of Andrology* 7, no. 3 (2005): 257–262, https://doi.org/10.1111/j.1745-7262.2005.00047.x.

57. F. Kong, W. Du, Y. Liang, et al., "Exploring Ginsenoside Rg1 in Improving Fertility and Mitochondrial Function," *Journal of Ethnopharmacology* 295 (2022): 115421, https://doi.org/10.1016/j.jep.2022.115421.

58. Z. Liu, W. Jiang, M. Wen, et al., "Network Pharmacology and Experimental Validation to Explore the Mechanism of *Panax ginseng* in Treating Premature Ovarian Insufficiency," *Frontiers in Pharmacology* 14 (2023): 1142786, https://doi.org/10.3389/fphar.2023.1142786.

59. G. Lin, X. Li, S. L. J. Yie, and L. Xu, "Clinical Evidence of Coenzyme Q10 Pretreatment for Women With Diminished Ovarian Reserve Undergoing IVF/ICSI: A Systematic Review and Meta-Analysis," *Annals of Medicine* 56, no. 1 (2024): 2389469, https://doi.org/10.1080/07853890.2024.2389469.

60. M. P. Murphy and R. C. Hartley, "Mitochondria as a Therapeutic Target for Common Pathologies," *Nature Reviews Drug Discovery* 17, no. 12 (2018): 865–886.

61. M. P. Murphy and R. A. J. Smith, "Targeting Antioxidants to Mitochondria by Conjugation to Lipophilic Cations," *Annual Review of Pharmacology and Toxicology* 47 (2007): 629–656.

62. J. Zielonka, J. Joseph, A. Sikora, et al., "Mitochondria-Targeted Triphenylphosphonium-Based Compounds: Syntheses, Mechanisms of Action, and Therapeutic and Diagnostic Applications," *Chemical Reviews* 117, no. 15 (2017): 10043–10120.

63. D. Kalyane, N. Raval, R. Maheshwari, V. Tambe, K. Kalia, and R. K. Tekade, "Employment of Enhanced Permeability and Retention Effect for Improved Nanoparticulate Drug Delivery Systems in Tumor Targeting," *Drug Discovery Today* 24, no. 3 (2019): 781–796.

64. G. Battogtokh, Y. S. Choi, D. S. Kang, et al., "Mitochondria-Targeting Drug Conjugates for Cytotoxic, Anti-Oxidizing and Sensing Purposes: Current Strategies and Future Perspectives," *Acta Pharmaceutica Sinica B* 8, no. 6 (2018): 862–880.

65. S. Biswas, N. S. Dodwadkar, P. P. Deshpande, and V. P. Torchilin, "Liposomes Loaded With Paclitaxel and Modified With Novel Triphenylphosphonium-PEG-PE Conjugate Possess Low Toxicity, Target Mitochondria and Demonstrate Enhanced Antitumor Effects In Vitro and In Vivo," *Journal of Controlled Release* 159, no. 3 (2014): 393–402.

66. D. M. Pegtel and S. J. Gould, "Exosomes," *Annual Review of Biochemistry* 88 (2019): 487–514.

67. B. Sun, Y. Ma, F. Wang, L. Hu, Y. Sun, and Z. Wang, "Exosomes From Human Umbilical Cord Mesenchymal Stem Cells Restore Ovarian Function in Chemotherapy-Induced Premature Ovarian Insufficiency Mice," *Stem Cell Research & Therapy* 8 (2017): 10.

68. M. Yang, L. Lin, C. Sha, et al., "Bone Marrow Mesenchymal Stem Cell-Derived Exosomes Improve Ovarian Function and Proliferation in Premature Ovarian Insufficiency by Regulating Mitochondrial Function and Apoptosis," *Stem Cell Research & Therapy* 11 (2020): 440.

69. A. F. Makled, I. M. El-Sherbiny, and M. El-Sherbiny, "Ovarian-Targeted Nanomedicine: Emerging Opportunities for Infertility Management and Ovarian Rejuvenation," *International Journal of Pharmaceutics* 641 (2023): 123065.

70. A. Ochiai and K. Kuroda, "Preconception Resveratrol Intake Against Infertility: Friend or Foe?," *Reproductive Medicine and Biology* 19, no. 2 (2020): 107–113, https://doi.org/10.1002/rmb2.12303.

71. EFSA Panel on Food Additives and Nutrient Sources Added to Food (ANS), "Scientific Opinion on the Safety of Green Tea Catechins," *EFSA Journal* 16, no. 4 (2018): e05239, https://doi.org/10.2903/j.efsa.2018.5239.

72. S. Andres, S. Pevny, R. Ziegenhagen, et al., "Safety Aspects of the Use of Quercetin as a Dietary Supplement," *Molecular Nutrition & Food Research* 62, no. 1 (2018): 1700447, https://doi.org/10.1002/mnfr.201700447.

73. E. Chan, "Displacement of Bilirubin From Albumin by Berberine," *Biology of the Neonate* 63, no. 4 (1993): 201–208, https://doi.org/10.1159/000243934.

74. C. S. Yuan, G. Wei, L. Dey, et al., "Brief Communication: American Ginseng Reduces Warfarin's Effect in Healthy Patients: A Randomized, Controlled Trial," *Annals of Internal Medicine* 141, no. 1 (2004): 23–27, https://doi.org/10.7326/0003-4819-141-1-200407060-00011.

75. T. Hidaka, K. Fujii, I. Funahashi, N. Fukutomi, and K. Hosoe, "Safety Assessment of Coenzyme Q10 (CoQ10)," *BioFactors* 32, no. 1–4 (2008): 199–208, https://doi.org/10.1002/biof.5520320124.

76. V. Desquiret-Dumas, A. Clément, V. Seegers, et al., "The Mitochondrial DNA Content of Cumulus Granulosa Cells Is Linked to Embryo Quality," *Human Reproduction* 32, no. 3 (2017): 607–614, https://doi.org/10.1093/humrep/dew341.

14470578, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rmb2.70086, Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License